

Limitations of the Sequential Parallel Comparisons Design

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Outline

I. The Design

II. Estimands & Estimators

II. Assumptions

III. Misclassification

IV. Simulation

V. Discussion

Diagram

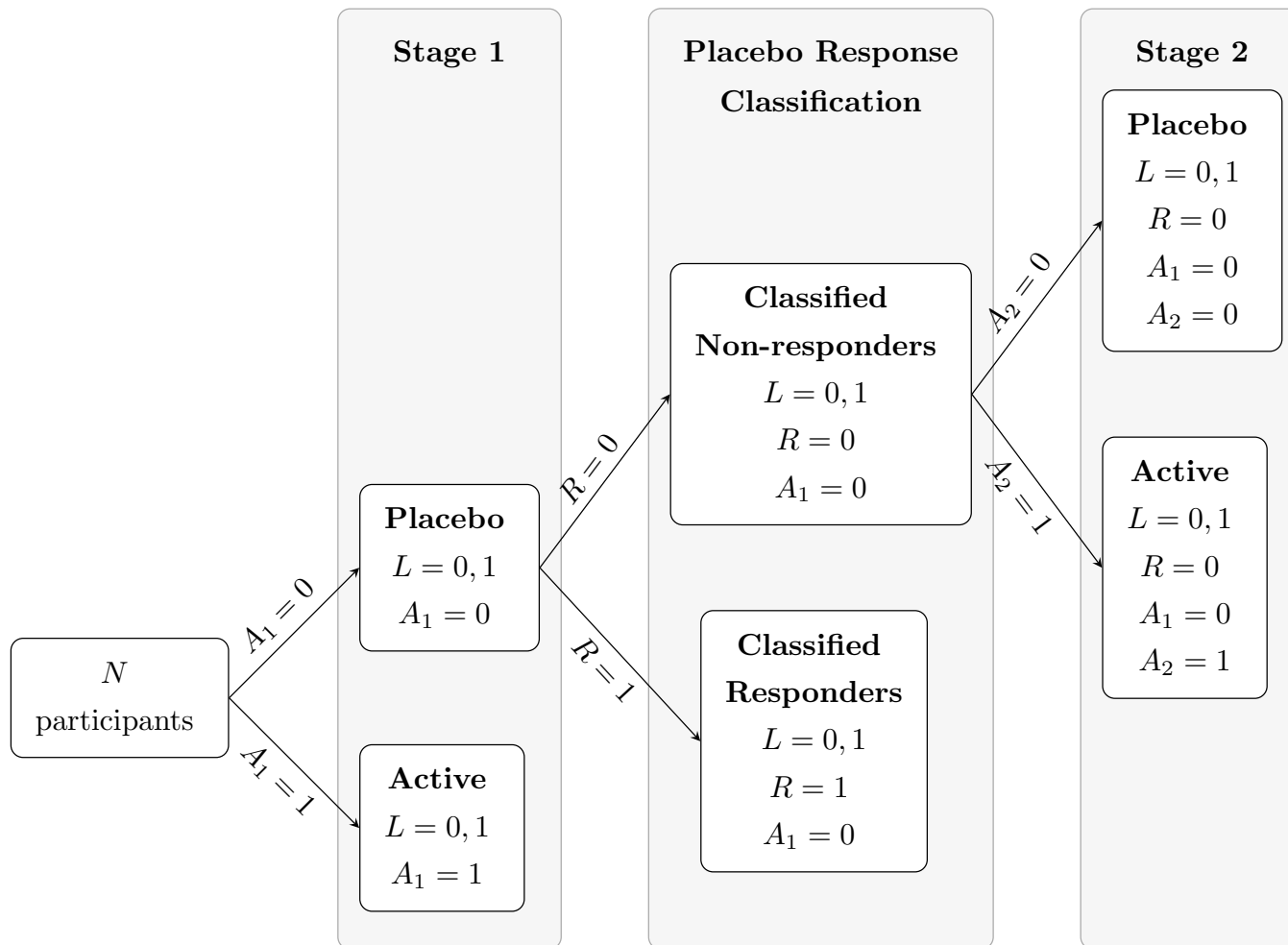


Figure 1: An overview of the Sequential Parallel Comparisons Design (SPCD) [1].

Example Data

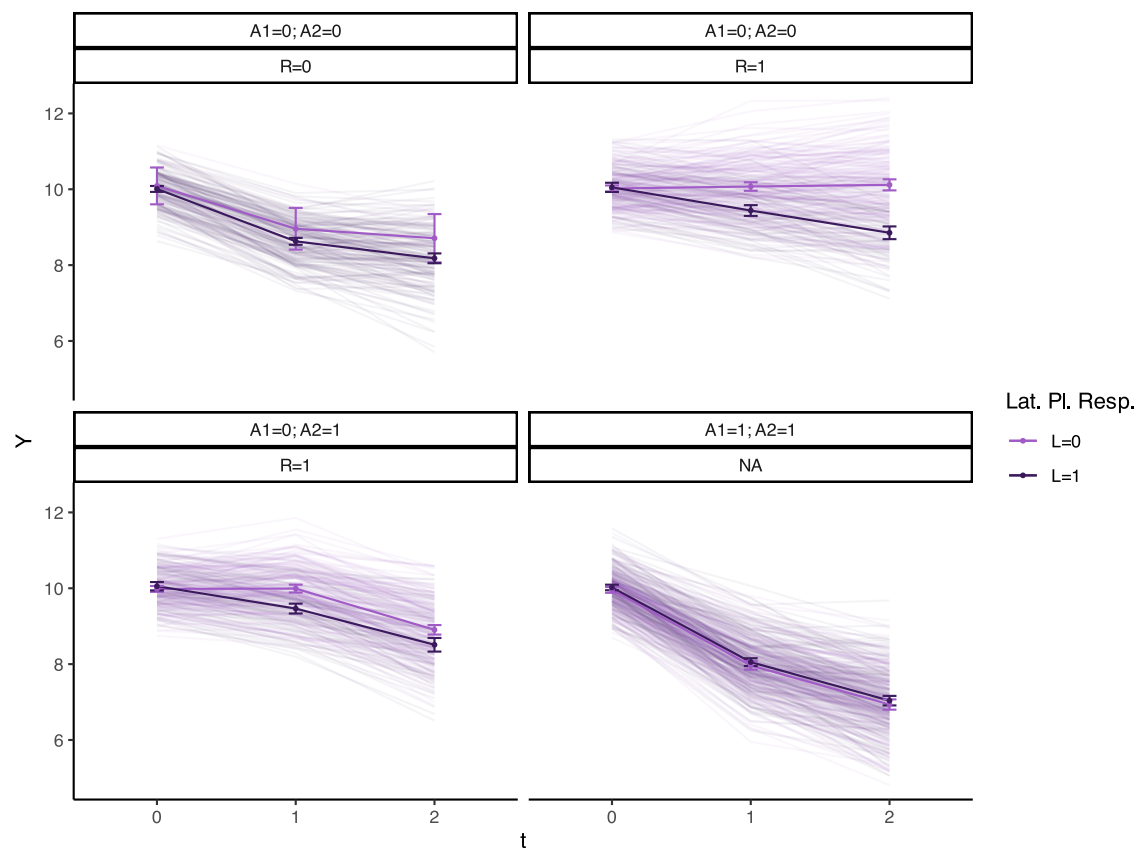


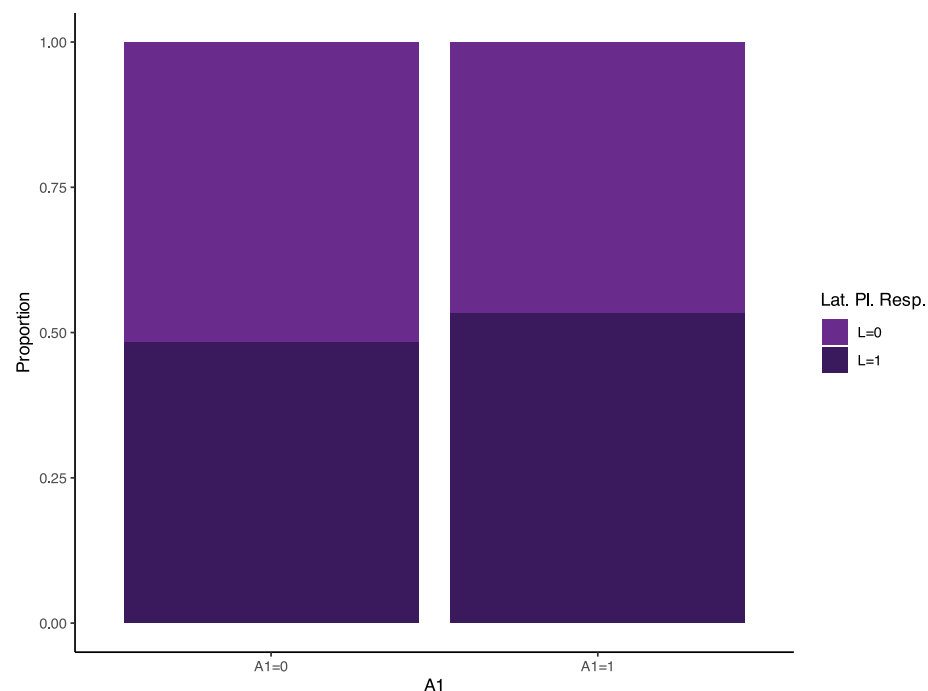
Figure 2: Spaghetti plot of simulated SPCD data color-coded by latent placebo response status (L).

Stage 1

- Analogous to a Parallel Arm Randomized Control Trial
- Placebo treated arm is a latent mixture of placebo responders ($L = 1$) and non-responders ($L = 0$)
- Natural contrast is between Active ($A_1 = 1$) and Placebo ($A_1 = 0$) Treatments

$$Y_1(A_1 = 1) - Y_1(A_1 = 0)$$

$$\hat{\theta}_1 = \hat{E}(Y_1(1) \mid Y_0) - \hat{E}(Y_1(0) \mid Y_0)$$



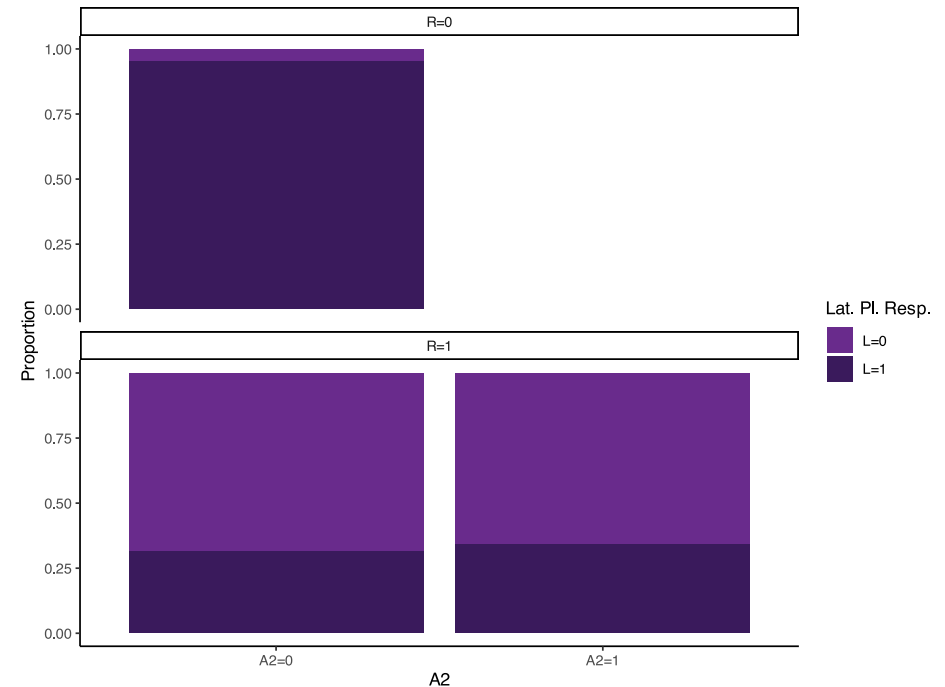
Mixture of latent placebo response status (L) in Stage 1 treatment arms.

Stage 2

- After Stage 1, placebo participants are classified as responders ($R = 1$) or non-responders ($R = 0$)
- **Classified** non-responders will inevitably contain some **true** placebo responders due to *misclassification*
- Natural contrast is between Active and Placebo for classified placebo non-responders

$$Y_2(A_2 = 1) - Y_2(A_2 = 0) \mid R = 0, A_1 = 0$$

$$\begin{aligned} \hat{\theta}_2 = & \hat{E}(Y_2(1) \mid R = 0, A_1 = 0, Y_1, Y_0) \\ & - \hat{E}(Y_2(0) \mid R = 0, A_1 = 0, Y_1, Y_0) \end{aligned}$$



Mixture of latent placebo response status (L) in Stage 2 treatment arms sorted by classified response status.

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Estimands

Quantity	Population Subset	Formula
Δ_{placebo}	Stage 1 Placebo; $A_1 = 0$	$E(Y_1(0) \mid L = 1) - E(Y_1(0) \mid L = 0)$
Δ_{NR}	Placebo NR; $L = 0$	$\mathbb{E}(Y_1(1) \mid L = 0) - E(Y_1(0) \mid L = 0)$
Δ_{PR}	Placebo Resp.; $L = 1$	$\Delta_{NR} - \Delta_{\text{placebo}}$
Δ_{all}	All Participants	$\Delta_{NR} - P(L = 0)\Delta_{\text{placebo}}$

Table 1: Average effects as functions of counterfactual outcomes and latent placebo responder status.

Estimators

Estimator	Targeted Estimand	Formula
$\hat{\theta}_1$	Δ_{all}	$\hat{E}(Y_1(1) Y_0) - \hat{E}(Y_1(0) Y_0)$
$\hat{\theta}_2$	Δ_{NR}	$\hat{E}(Y_2(1) R = 0, Y_1, Y_0) - \hat{E}(Y_2(0) R = 0, Y_1, Y_0)$
$\hat{\theta}_w$	?	$w\hat{\theta}_1 + (1 - w)\hat{\theta}_2$

Table 2: Conventional estimators for the SPCD.

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Conventional Identification Assumptions

Consistency/SUTVA

$$Y_1 = Y_1(A_1) \mid L \quad \text{and} \quad Y_{i,1}(A_{i,1} = a) \perp A_{j,1} \mid L = l \quad \text{for all } i \neq j.$$

Ignorability

$$A_1 \perp (Y_1(1), Y_1(1), Y_1(0), Y_1(0))'$$

Positivity

$$0 < P(A_1 = a) < 1 \quad \text{for all } a.$$

Implausible, but Required Assumptions

Latent-type Stability

$$L \perp A_1$$

Effect Homogeneity

$$E(Y_t(A_t = 1) \mid L = 1) = E(Y_t(A_t = 1) \mid L = 0)$$

Mixture Identifiability

Given that Y_1 follows a mixture, let $p_L = P(L = 1)$ be the unknown placebo responder prevalence, $f_{Y_1}(\cdot; \xi)$ be component densities from family $\mathcal{F} = \{f(\cdot; \xi)\}$ indexed by ξ_l for latent response status $L = l$. Then the density of the stage 1 outcome in the placebo arm is,

$$f_{Y_1|A_1=0}(y) = p_L f_{Y_1}(y; \xi_1) + (1 - p_L) f_{Y_1}(y; \xi_0),$$

such that the mixture is identified under that parametric model.

Estimator Expectations

Estimator	Targeted Estimand	Actual Expectation
$\hat{\theta}_1$	Δ_{all}	Δ_{all}
$\hat{\theta}_2$	Δ_{NR}	$\Delta_{NR} - p_{L \mid R=0} \Delta_{\text{placebo}}$
$\hat{\theta}_w$	?	$\Delta_{NR} - \Delta_{\text{placebo}}[wp_L + (1 - w)p_{L \mid R=0}]$

Table 3: Expectations of the conventional estimators. Note, $p_L = P(L = 1)$ and $p_{L \mid R=0} = P(L = 1 \mid R = 0)$.

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Misclassification is Inevitable...

- A limitation of previous work on SPCD is the absence of substantial discussion of placebo response classification, with some recent exceptions[2], [3], [4], [5].

$$C_R(y_0, y_1) = I(y_1 - y_0 > c) \text{ or } C_R(y_0, y_1) = I(y_1 > c), \quad (1)$$

- If placebo responders are misclassified as non-responders, then the estimate for Δ_{NR} is contaminated
- Responder misclassification is inevitable unless $C_R(y_0, y_1) = 1$ for all participants (no non-responders)

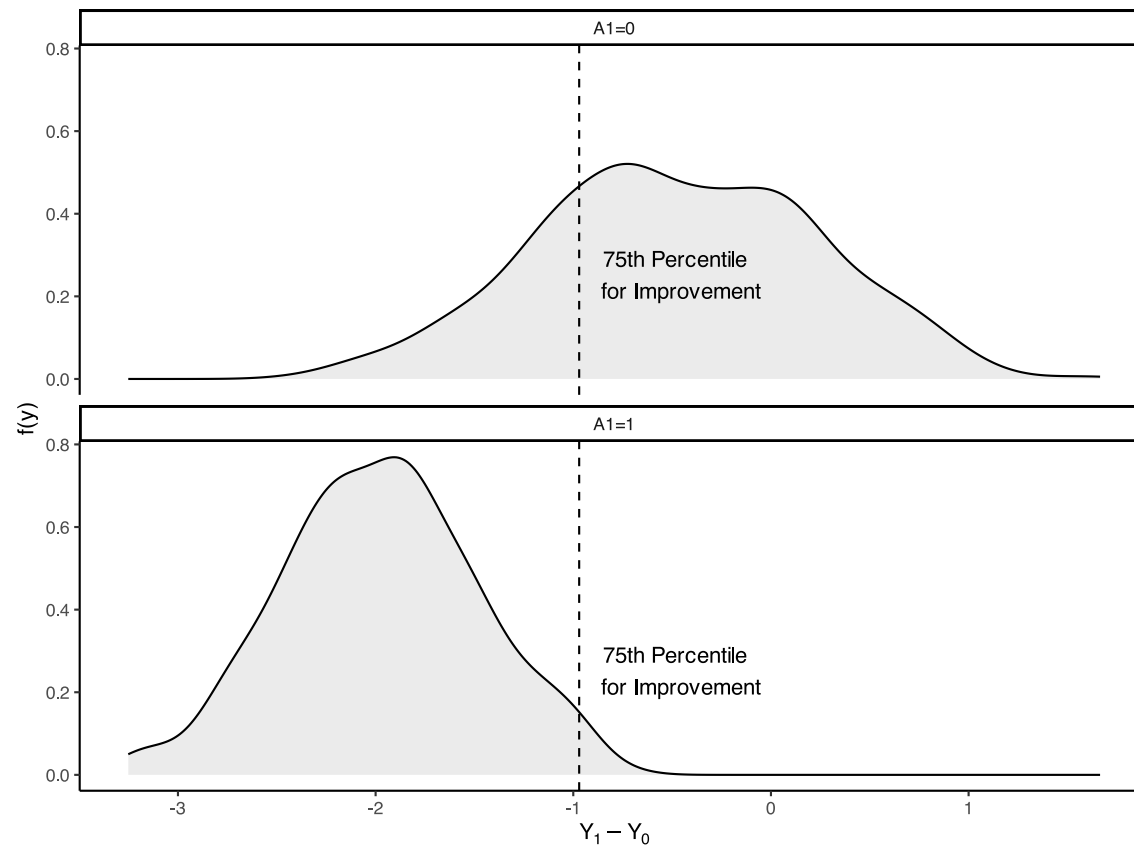


Figure 3: Distributions of individual responses after stage 1.

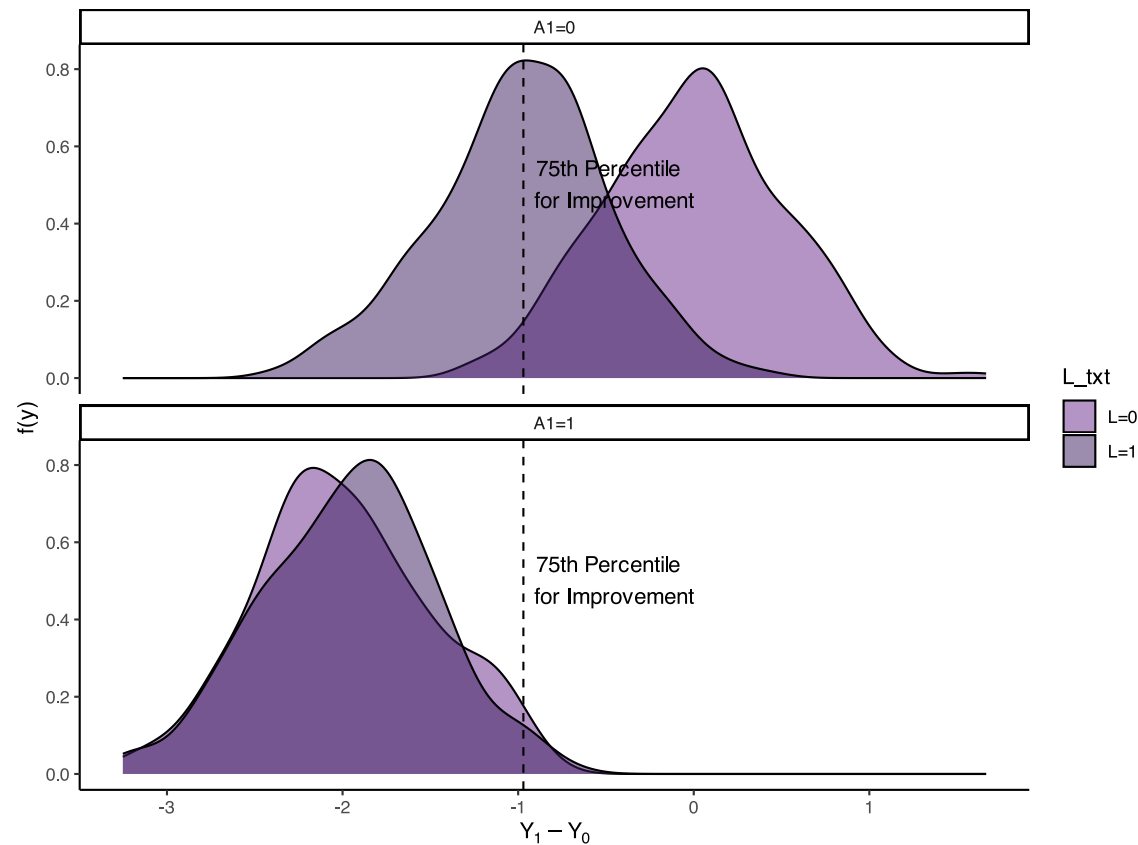


Figure 4: Distributions of individual responses after stage 1 by Stage 1 Treatment & Latent placebo response status.

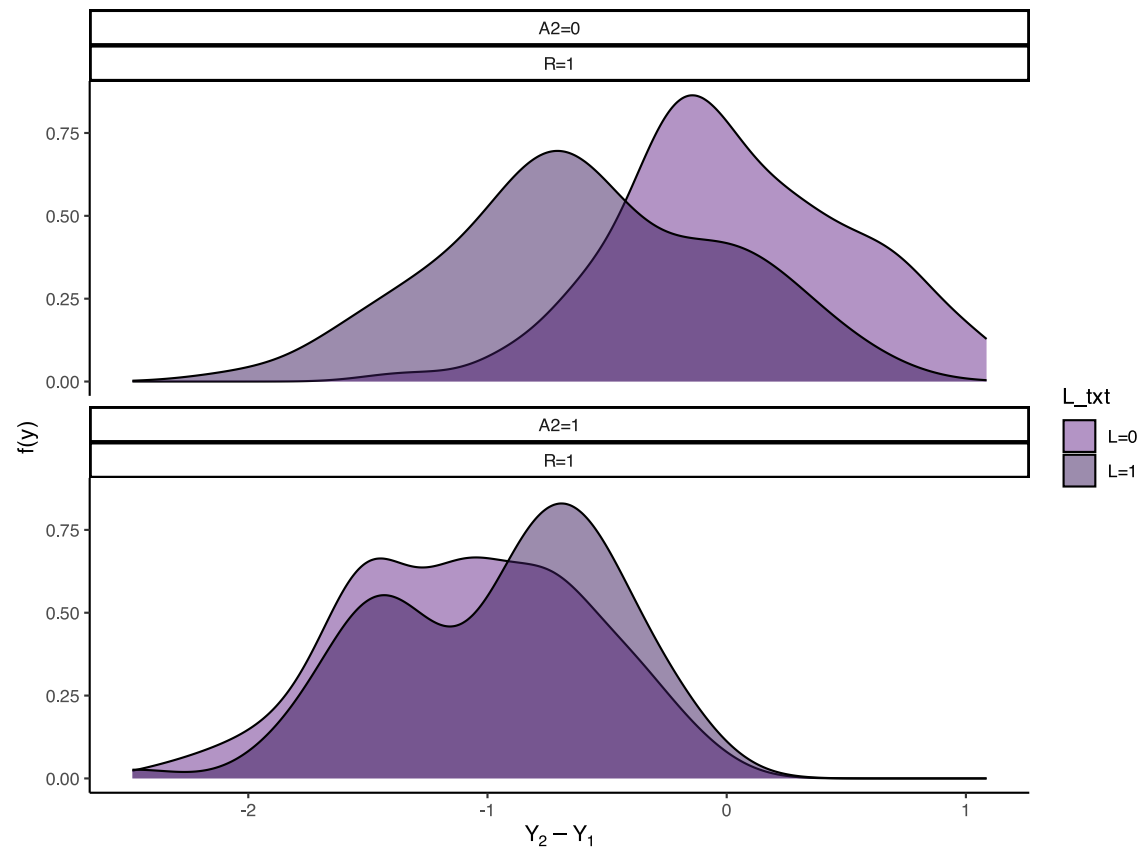


Figure 5: Distributions of individual responses after stage 2.

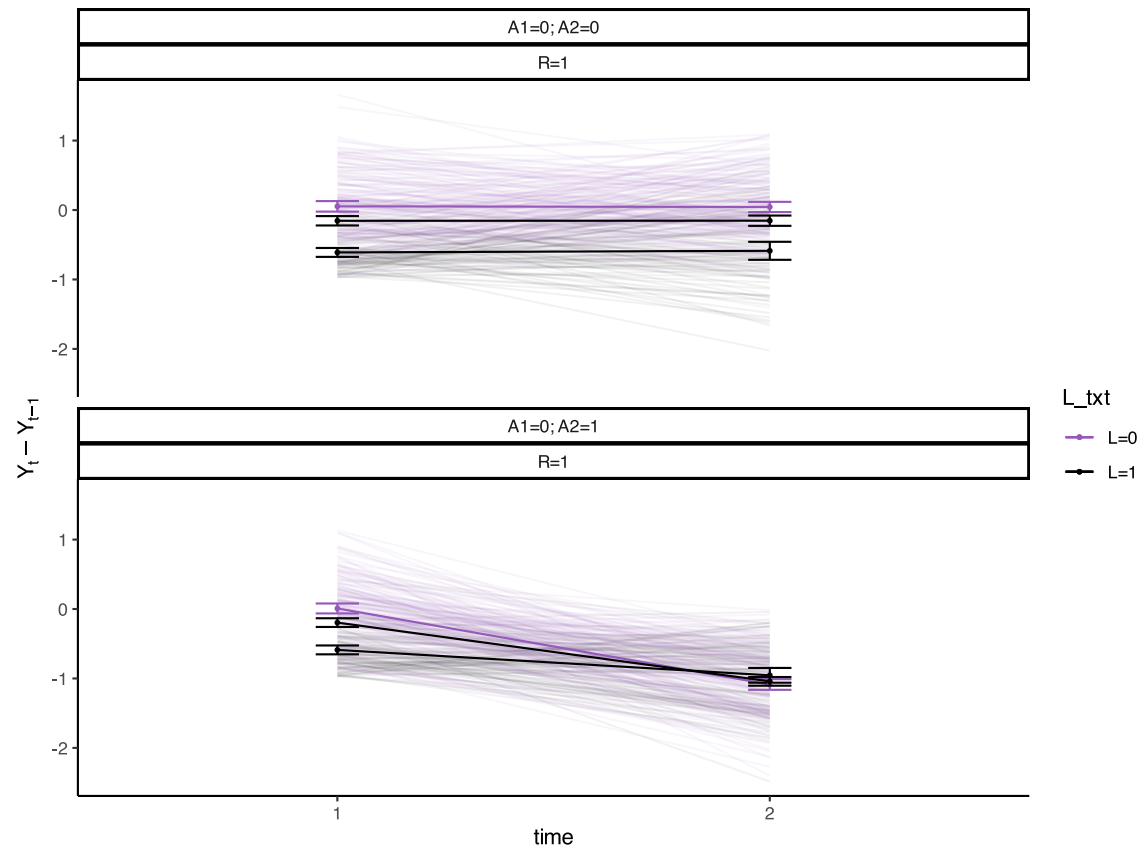


Figure 6: Effect estimates

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Simulation Results

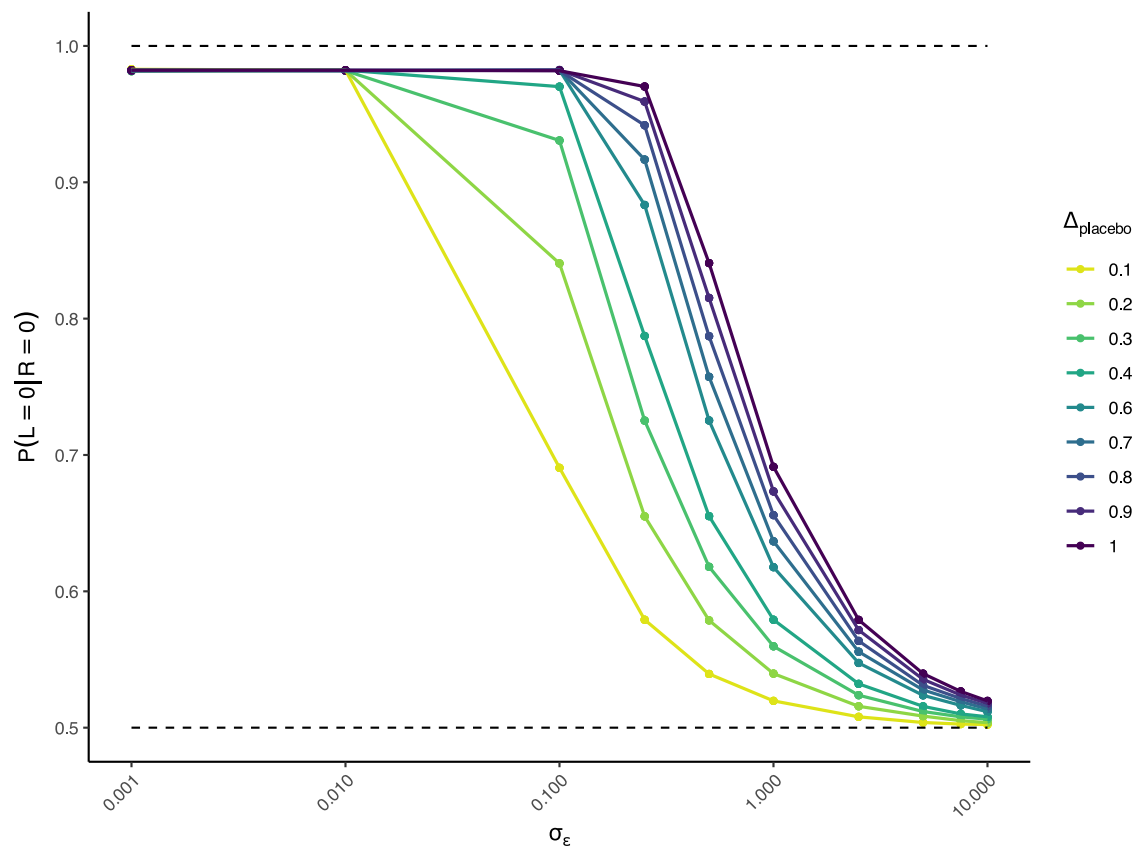


Figure 7: The average negative predictive values $P(L = 0 \mid R = 0)$ for the threshold classifier used in the simulation study.

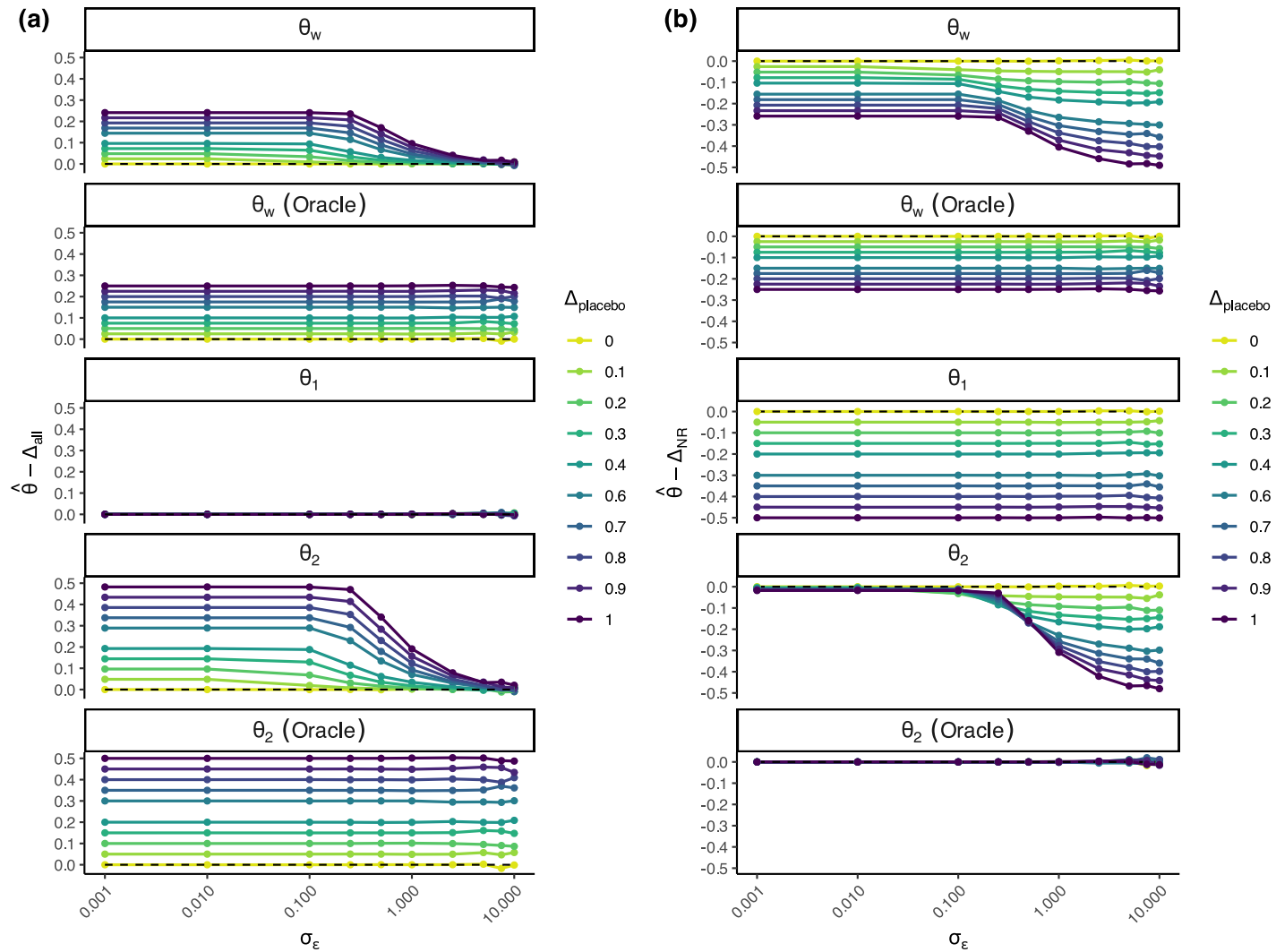


Figure 8: Simulation under the null; $\Delta_{\text{all}} = 0$, with varied Δ_{placebo} and residual standard deviation σ_σ .

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Take-aways

- SPCD has several implausible assumptions baked into the design
- Misclassification makes the estimation unreliable even if you are willing to make the assumptions
- The conventional Stage 2 and weighted estimands don't target anything meaningful and are difficult to interpret

Warning

Don't use the SPCD!

Alternative...

... approaches to placebo response:

- Latent regression with a continuous placebo
- Optimal partitioning via finite mixture models

... designs:

- Placebo lead-in
- Conventional crossover

Acknowledgements

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Computation

The computational work performed on this project was done with help from the NYU Big Purple High-Performance Computing cluster.

Questions?

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